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Adaptive Neuro Fuzzy Inference System based classifier in diagnosis of breast cancer

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ABSTRACT

Millions of individuals live longer and healthier lives due to modern medical innovations. The advancement of modern medicine to combat numerous ailments has benefited humanity a lot. Breast cancer is one of the most fatal diseases affecting women globally. Early detection is essential for lowering the fatality rate. The objective of the paper is to create a computer-aided diagnostic model that helps in the early detection of breast cancer and hence decreases the death rate. The study introduces a hybrid strategy for effectively diagnosing breast cancer by using a Novel Relief algorithm for feature selection with an Adaptive Neuro-Fuzzy Inference System (ANFIS). The efficiency of this proposed hybrid model and the ANFIS model without using any feature selection technique is estimated using the Wisconsin Breast Cancer Data set (WBCD). The study finds that the new hybrid model has attained the highest accuracy of 99.30% and is ideal for detecting breast cancer.

1. Introduction

In any disease, it is very much essential to make correct diagnosis at the initial stage without much delay. There are many diseases present in the real life, if they are not identified at an early stage and not cured soon then they can lead to many deaths. Over the time, Artificial Intelligence (AI) is proving very useful in medical diagnosis. AI has been developed using Machine Learning (ML) techniques. There are numerous developed algorithms, methods, models, software, etc., to recognize variables and patterns of the various diseases from which machine creates their logic to ascertain the diagnosis. Identifying and curing diseases early will also lower the mortality rate. To acquire essential and beneficial resourcefulness, a tremendous amount of data is trained and tested simultaneously using various techniques of ML.

Soft computing is one of the major part of ML. Several soft computing algorithm like Artificial Neural Network (ANN), Support Vector Machine (SVM), Fuzzy-Logic, Adaptive Fuzzy Inference System (ANFIS), Genetic algorithms, etc., are widely used in detecting breast cancer before many clinical investigations.

Feature selection also plays a vital role in ML. ML starts by adding as many features as possible to achieve more reliable outcomes. However, the model's performance degrades as the number of features increases, also known as "The Curse of Dimensionality". Due to this system, they do not behave appropriately and accurately, which is a significant challenge. To deal with such a problem, we propose a Novel Relief algorithm for feature selection.

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Breast cancer is the second-most widespread invasive cancer worldwide among women. In 2020, the World Health Organization reported that breast cancer was diagnosed in 2.3 million women worldwide, resulting in 685,000 deaths [https://www.who.int/news-room/fact-sheets/detail/breast-cancer]. Mortality in women due to breast cancer is around 1 in 38, i.e., 2.6%. ACS also surveyed that out of 268,600 cancerous women, 41,670 women died due to breast cancer [https://www.medicalnewstoday.com/articles/37136]. Breast cancer is a dangerous disease, as one-third of women die from breast cancer. If it is curable at an early stage, the death rate can be decreased.

According to the article by L.A. Zadeh [1], "Knowledge-based systems are based on fuzzy logic, which has been applied in many fields like home appliances, automobiles, control, and medicine". Fuzzy Logic employs its own lexicon, such as Fuzzification, Defuzzification, Membership Function, Linguistic variables, Domains, Rules, and so on [1]. The Fuzzy Logic Control System (FLC) develops strategies to handle forbidden operations, which are at the core of human intelligence, knowledge, and understanding. However, expert systems are unable to manage complex processes, while FLC systems are utilized for ambiguous processes (ref. Faran B. et al.) [2].

The Adaptive Neuro-Fuzzy Inference System (ANFIS) is a computational model that combines the adaptive capabilities of neural networks with the interpretability of fuzzy logic. The motivation behind using ANFIS lies in addressing certain challenges and leveraging the strengths of both fuzzy systems and neural networks, such as handling uncertainty in Fuzzy logic and ANN, knowledge representation, adaptability, hybrid modeling, universal function approximation, and reducing expert dependency. The hybrid nature of ANFIS allows it to offer a balanced and powerful approach to modeling complex systems. ANN has learning capabilities, but it cannot interpret results. That is, it acts as a black box. Whereas the Fuzzy Inference System (FIS) can interpret the results using a rule base, it cannot learn. To overcome the disadvantages of both techniques, joint use of ANN and FIS can be employed to get better results.

1.1. Contribution

In this paper, we modify the existing Relief algorithm for feature selection by eliminating outliers, imputing missing values, and using the Mahalanobis distance technique. We comprehend this modified Relief algorithm as a Novel Relief algorithm. We propose the ANFIS model with two approaches: one that uses a Novel Relief algorithm for feature selection technique and the other without any feature selection technique. Both the ANFIS models have been tested on the WBCD and classification accuracy has been obtained. The hybrid simulation of the FIS with ANN and the Novel Relief algorithm gives the highest classification accuracy. This paper explores several important results of the proposed ANFIS model that diagnoses breast cancer. These proposed approaches are useful in the development of a diagnostic system for breast cancer. The proposed new hybrid ANFIS model can be used as a diagnostic system and can assist practitioners in making quick diagnoses.

The paper is organized as follows: Section 2 represents the literature survey. Section 3 briefly introduces the Adaptive Neuro-Fuzzy Inference System (ANFIS), a Novel Relief algorithm and details of the experimental data. Section 4 shows a comparative analysis and explanation of the results obtained. The conclusion is included in Section 5.

2. Literature survey

P.R. Innocent et al. [3] conducted a study of fuzzy methods for medical diagnosis in nursing assessment using Type-II fuzzy sets (2007). Faran Blag et al. [2] designed a control system using fuzzy logic for the normality of human function in the human brain and also made a medical diagnosis of brain tumor and hemorrhage (2011). Manish Rana et al. [4] proposed an expert system using fuzzy logic to diagnose hemorrhage, brain tumor, cardiac disease, and thyroid (2013). J.B. Awotunde et al. [4] proposed a medical diagnosis system using fuzzy logic for malaria disease (2014). M.A. Madkour et al. [5] developed a model using Fuzzy logic for the diagnosis of flu. They had also implemented this model for common Measles, German measles, Mumps, Chickenpox, Whooping cough, Common cold and Meningitis.

Elif Derya Ubeyli [6] proposed an integrated view of ANFIS to detect Breast Cancer and tested the model on the WBCD data set (2008). Seyedesh S.N. et al. [7] designed a hierarchical fuzzy neural system with Extended Kalman Filter (EKF). They applied this model to the WBCD data set. The model is compared with the Hierarchical Fuzzy Neural System (HFFN)+ EKF and FNN and obtained the results (2010). Gerald S. et al. [8] proposed a hybrid cost-sensitive fuzzy classification for Breast Cancer diagnosis. They investigated with Michigan, and Diffsburgh-style approaches with a hybrid GA-ANFIS classifier (2010). Somayesh N. et al. [9] proposed a classification model for Breast Cancer based on advanced multi-dimensional fuzzy NN. They applied Hierarchical FNN + Fuzzy Gaussian Potential NN to the WBCD data set with the new training algorithm HFNN, which uses fewer rules and parameters to model the nonlinear system. In the antecedent, they used the Gaussian Potential Function as a membership function (MF) (2011). Shweta saxena et al. [10] surveyed different machine learning algorithms like ANFIS, SVM, and fuzzy ANFIS and obtained comparative results with other author work (2012). Manisha Arora et al. [11] experimented with a neuro-fuzzy expert system for the WBCD dataset with different MFs and a Sugeno–Mamdani fuzzy inference system in MATLAB (2012). Fatima et al. [12] experimented with a neuro-fuzzy model on the WBCD dataset for the recognition of breast cancer. Payam et al. [13] proposed an optimized ANFIS model with a feature selection method for detecting breast cancer at an early stage. In the study, they used the Association Rules (AR) method to select features from the dataset. In the proposed model, the Cuckoo optimization algorithm is used to learn ANFIS and has obtained high classification accuracy.

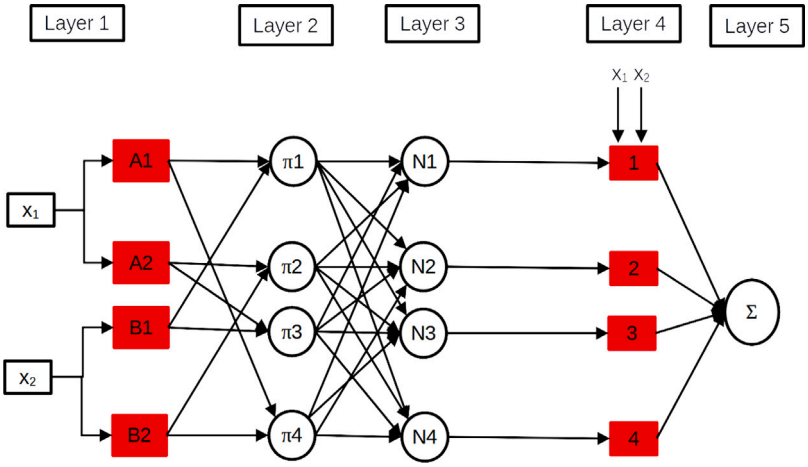


Fig. 1. Architecture of ANFIS model with two inputs, one output, and two rules.

Ejiofor et al. [14] used the ANFIS model for the diagnosis of Breast Cancer using MATLAB. A Sakthviel et al. [13] compared SVM and ANFIS for breast cancer detection. In the study, they used the Rough and Genetic algorithm as a feature selection technique. The classification accuracy of both techniques is compared. Indu Gola et al. [15] implemented fuzzy classification with a comprehensive learning Gravitational search algorithm in Brain Tumor detection. Also, they used 10-fold cross-validation to split the train–test dataset. Ahmed Rizal et al. [16] developed ANFIS with subtractive clustering for breast cancer detection and reduced the error of classification. Amany M.L. et al. [17] proposed two approaches to the diagnosis of Breast Cancer. In the first approach, they used the evolution Genetic Algorithm with ANFIS and obtained results. In the second approach, ANFIS is implemented with Principal Component Analysis as a feature selection technique, obtaining comparative results. We compare both proposed models for accuracy. Indira Mahic [16] introduced a new approach to Breast Cancer diagnosis using a Fuzzy c-means algorithm and pattern recognition method. This method was applied to Breast Cancer climate samples and obtained high accuracy. M. Ashraf et al. [18] introduced an information-gain technique with ANFIS for Breast Cancer diagnosis. Hazlina et al. [19] presented the ANFIS model for Breast Cancer survival. In these studies, they used a partial logistic ANN model to predict the hazard curve and survival wave of Breast Cancer patients (2014). Wahyani et al. [20] presented a comparative study on fuzzy Mamdami–Sugno Tsukamoto for the children's tuberculosis diagnosis.

3. Methodology

3.1. Adaptive Neuro-Fuzzy Inference System (ANFIS)

The ANFIS was developed by Jyh-Shing Roger Jang in 1993 [21]. An Adaptive Neuro-Fuzzy Inference System is a hybrid of Artificial Neural Network and Fuzzy Inference system, also known as an adaptive network-based fuzzy inference system. ANFIS is a set of algorithms for ANN that is modeled on the Takagi–Sugeno Fuzzy Inference System.

A multi-layer feed-forward network made up of nodes connected by directed interconnections forms an adaptive network. Each node executes the specific function on its receiving signals to produce a signal node output [21,22]. Each interconnection in the adaptive network describes the direction of signal flow from one node to another without assigning any weight to the network. The adaptive network runs on two types of nodes, i.e., static node and adaptive node.

Fig. 1 represents the architecture of the ANFIS model with two inputs, one output, and two rules. The overall system design is made up of five layers, namely: (i) Fuzzification layer, (ii) Rule layer, (iii) Normalized layer, (iv) Defuzzification layer, (v) Overall output (or summation neuron).

A basic Takagi–Sugeno fuzzy rule is describe by:

$$IF\ (x_1\ is\ A_1)\ AND\ (x_2\ is\ A_2)\ \ldots\ AND\ (x_m\ is\ A_m)\ THEN\ y = f(x_1, x_2, \ldots, x_m)$$

Where, $x_1, x_2, \ldots, x_m$ are input variables $A_1, A_2, \ldots, A_m$ are fuzzy sets and y is either a constant or linear function of input variables [21,23].

Consider the first order, Takagi–Sugeno FIS, for two rules [22,24].

$$Rule\ 1: IF\ (x_1\ is\ A_1)\ AND\ (x_2\ is\ B_1)\ THEN\ y_2 = f_1 = k_{10} + k_{11}x_1 + k_{12}x_2$$

$$Rule\ 2: IF\ (x_1\ is\ A_2)\ AND\ (x_2\ is\ B_2)\ THEN\ y_2 = f_2 = k_{20} + k_{21}x_1 + k_{22}x_2$$

Where, x_1 and x_2 are input variables. A_i and B_i are fuzzy sets that represent the membership function of the ANFIS antecedent. k_{i0}, k_{i1}, and k_{i2} are linear consequent parameters that are specified for each rule i. As shown in Fig. 1, a circle indicates the static or fixed node, and a square indicates the adaptive node, i.e. during adaptation or training, the parameter can be novel. A brief overview of the ANFIS algorithm is explained as follows [22,24]:

Layer 1: - Fuzzification layer:

In this layer, each crisp input is fuzzified using the membership function. All the nodes in this layer are adaptive nodes.

$$O_i^1 = \mu_{A_i}(X); i = 1,2 \tag{1}$$

$$O_i^1 = \mu_{B_j}(X); j = 1,2 \tag{2}$$

Where, $\mu_{A_i}(X)$ and $\mu_{B_j}(X)$ can be determined using any membership function. Let Gaussian membership function is employed to fuzzify the input variables.

The Gaussian membership function is given by,

$$\mu_{A_i}(x_1) = exp\left\{-\left(\frac{x_1 - a_i}{b_i}\right)^2\right\}; i = 1,2 \tag{3}$$

Where a_i and b_i are parameters of membership functions.

Layer 2: - Rule layer:

The nodes in this layer are fixed nodes. The fixed nodes in this layer are labeled with π. Each neuron of this layer is associated with a particular Sugeno-FIS and calculates the firing strength or weight of the rules, respectively. This layer obtains the output as follows:

$$O_i^2 = w_i = \mu_{A_i}(x_1)\mu_{B_j}(x_2);\ i = 1,2 \tag{4}$$

where, w_i represents the firing strength of the ith rule.

Layer 3: - Normalized layer:

Nodes in this layer are also fixed nodes. In this layer, each node represents the normalized firing strength that was obtained in the previous layer, and it is indicated by N, as shown in Fig. 1. The normalization of the firing strength is found using the following equation.

$$O_i^3 = \bar{w_l} = \frac{w_i}{w_1 + w_2};\ i = 1,2 \tag{5}$$

Layer 4: - Defuzzification layer:

The nodes in this layer are adaptive nodes. Weighted average defuzzification obtains the output of this layer. The nodes receive initial inputs x_1 and x_2. The resulting output of this layer is computed using the following equation:

$$O_i^4 = \bar{w_l} f_i = \bar{w_l}(k_{i0} + k_{i1}x_1 + k_{i2}x_2);\ i = 1,2 \tag{6}$$

Layer 5: - Overall output:

This layer is referred to as the output layer since it aggregates all the nodes of the previous layer 4 and converts fuzzy classification results into crisp values. This layer has a single fixed node labeled as 'Σ'.

$$O_i^5 = \sum_{i=1}^{2} \bar{w_l} f_i = \frac{\sum_{i=1}^{2} w_i f_i}{w_1 + w_2} \tag{7}$$

The first and the fourth layer of ANFIS are adaptive layers. Parameters a_i and b_i of membership functions at layer one are adaptable, known as premise (or antecedent) parameters. Parameters k_{i0}, k_{i1}, k_{i2} of layer four are adaptive and known as consequent parameters.

3.1.1. The learning algorithm of ANFIS

Premise parameter namely a_i, b_i and consequent parameter k_{i0}, k_{i1}, k_{i2} are to be learned to calculate the output of ANFIS for given training data. The ANFIS output can be expressed as:

$$\begin{aligned} f &= \frac{w_1}{w_1 + w_2} f1 + \frac{w_2}{w_1 + w_2} f2 \\ &= \bar{w_1}(k_{10} + k_{11}x_1 + k_{12}x_2) + \bar{w_2}(k_{20} + k_{21}x_1 + k_{22}x_2) \\ &= \bar{w_1}k_{10} + (\bar{w_1}x_1)k_{11} + (\bar{w_1}x_2)k_{12} + (\bar{w_2})k_20 + (\bar{w_2}x_1)k_{21} + (\bar{w_2}x_2)k_{22} \end{aligned} \tag{8}$$

Eq. (8) represents the linear combination of adaptable consequent parameters k_{i0}, k_{i1} and k_{i2}. These consequent parameters are obtained using the least square method [24].

ANFIS employs a hybrid learning technique that combines the least-squares method with the gradient descent approach. ANFIS is composed of the forward pass and the backward pass, as shown in Table 1. Once optimal consequent parameters are obtained, premise parameters are adjusted using the Gradient Descent method.

Table 1
Two passes to learn parameter for ANFIS.

	Forward pass	Backward pass
Premise Parameter	Fixed	Gradient Decent method
Consequent Parameter	Least-Square method	Fixed

Table 2
Wisconsin Breast Cancer Data (WBCD) dataset.

Number	Attributes	Domain
1	Patient ID	–
2	Clump Thickness (CT)	1–10
3	Uniformity of cell size (UCS)	1–10
4	Uniformity of cell shape (UCH)	1–10
5	Marginal Adhesion (MA)	1–10
6	Single Epithelial cell size (SEC)	1–10
7	Bare Nuclei (BN)	1–10
8	Bland Chromatin (BC)	1–10
9	Normal Nucleoli (NN)	1–10
10	Mitoses (Mit)	1–10
11	Class	2 for Benign 4 for malignant

The fundamental goal of the ANFIS is to use a learning algorithm to obtain the optimal values of the parameters.

3.2. Data set description

In this work, Wisconsin Breast Cancer Data (WBCD) is used for analysis. This database was created from FNA (Fine Needle Aspirate) and collected by Dr. William H. Wolberg and Olvi L. Mangasarian at the University of Wisconsin–Madison hospital [25]. This data set is available on the UCI Machine-Learning repository [25]. The WBCD data set consists of 699 records, out of which 16 records contain missing values. The WBCD data set consists of records with 65.5% benign tumors and 34.5% malignant tumors. Data set consists of 11 attributes, including the patient's ID and class labels, which are shown in Table 2. All features are scaled between 1–10.

3.3. Novel Relief algorithm

The Relief algorithm is a feature selection technique. The Relief algorithm was developed by Kira & Rendall in 1992 which was influenced by instance-based learning [26]. This algorithm is capable of dealing with both nominal and numerical variables. It cannot however cope with missing data and outliers and it is constrained to the two-class problem only. To overcome these problems, we propose a new distance-based learning Relief algorithm and comprehend it as a Novel Relief algorithm. The proposed Novel Relief algorithm deals with missing data and outliers. The Mahalanobis distance technique is used to select features. Missing values are obtained using Multivariate Imputation by Chained Equation (MICE) and Outliers are detected using Euclidean Distance. Outliers increase data uncertainty and reduce predictive capacity. As a consequence, eliminating outliers will increase the significance of the data.

3.3.1. Multivariate Imputation by Chained Equation (MICE)

1. Input all missing values with the mean of their respective columns as a starting point. Call it a 'zeroth' data set. Input the columns from left to right.
2. Remove the imputed values (from the 1st column from left), say x.
3. The remaining feature and rows become the feature matrix, and 'x_1' becomes the target variable. Apply the Linear Regression model to the rest of the columns '$x_1, x_2, x_3, \ldots, x_n$' to estimate the missing value in x_1 (as test data) and replace the missing value with the newly computed value. Repeat this procedure for those columns that have missing values.
4. Then subtract 'zeroth data set' with the latest ''transformed'' data set from which this new data set was formed. Repeat steps 2 and 3 with this unique data set until you get a stable model, i.e., until the difference between the two latest imputed data sets becomes very small.
5. Stop iterations when a pre-defined threshold is breached, or do it until a pre-defined maximum number of iterations are completed.

3.3.2. Novel Relief algorithm

Consider the training samples of size n, which are represented by $X = (x_1, x_2, \ldots, x_n)$. Each of these samples consists of a feature, which is given by $A = A_1, A_2, \ldots., A_a$. Each sample has a target value O_j. τ is the threshold parameter. The algorithm is briefly explained here.

Input:- n number of training samples having a vector of feature values and class values respectively.

Output:- The vector of W of feature scores which estimates the relevance of features.

1. Find missing values using MICE.
   Find outliers of data using Euclidean distance.
   Select m number of random training samples from n number of training samples ($n > m$).
2. Initialize all feature weights, say $W[A] = 0.0$
3. For $i = 1$ to m do begin
   Choose random 'target' instance R_i.
   Find nearest hit 'H' (of the same class value, say +ve) and nearest miss 'M' (of the opposite class value of the sample, say -ve) using Mahalonbis Distance.

   $$d(x, y) = \sqrt{(\vec{x} - \vec{\mu})^2 s^{-1} (\vec{x} - \vec{\mu})}$$

   where, $\mu \rightarrow$ mean and s $\rightarrow$ covariance matrix.
4. For each feature, update the weight:

   $$for A = 1 \; to \; a \; do$$

   $$W[A] = W[A] - diff(A, R_i, H_i) + diff(a, R_i, M_i)$$

   Where,

   $$diff(A, R_i, H_i) = \frac{\sqrt{(R_i - H_i)^2 s^{-1} (R_i - H_i)}}{max \; R_i - min \; R_i}$$

   and

   $$diff(A, R_i, M_i) = \frac{\sqrt{(R_i - M_i)^2 s^{-1} (R_i - M_i)}}{max \; R_i - min \; R_i}$$

5. Find relevance of feature:

   $$R_{f_i} = \frac{W}{m}$$

6. For $i = 1$ to a
   if $(R_{f_i} \geq \tau)$
   then The feature f_i is relevant
   else The feature f_i is irrelevant.
7. Print the vector W having feature performance with relevance.

In the second phase of this algorithm, we find the vector W having feature performance using Mahalanobis distance. The Novel Relief algorithm iteratively loops through m random training samples (T_{R_i}) out of n samples without repetitions. The weight of the feature can range from −1 to +1, where −1 indicates the worst score and +1 indicates the best score. Finally, a Novel Relief algorithm selects those features whose average weights are greater than the defined threshold.

4. Experiments

The performance of the Novel Relief-based ANFIS classifier is evaluated on the 'Wisconsin Breast Cancer Data set (WBCD)' for breast cancer classification. Four features, namely Clump Thickness, Uniformity of cell size, Marginal Adhesion and Mitoses, were found to be significant using the Novel Relief algorithm. Each feature consists of 3 membership functions and hence produces $3^4 = 81$ fuzzy rules.

The proposed ANFIS model with the Novel Relief algorithm is trained and tested as depicted in Fig. 2. In the first step, we input data into the ANFIS model. Then, in the second step, the data is normalized. Then a Novel Relief algorithm for feature selection is applied to the data set to reduce training time and obtain better precision with reduced over-fitting. As a part of the pre-processing, outliers of the data are found using Euclidean Distance and are neglected, moreover 16 missing values were computed using the MICE algorithm.

Fig. 3 represents the outliers of the WBCD data set, which interpret those areas that are far from the rest of the findings. As a consequence, the analysis performed on the WBCD data set may be biased. Hence, outliers detection is essential and is the first stage of the Novel Relief algorithm.

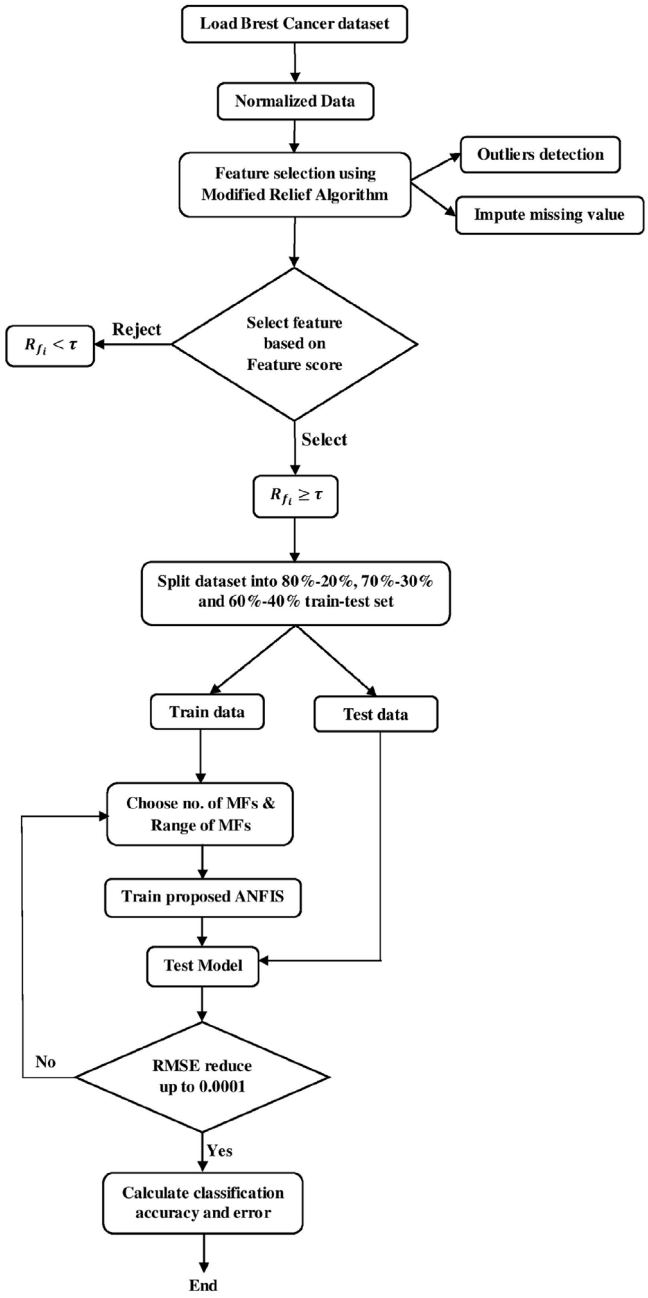


Fig. 2. Flowchart of proposed ANFIS model.

In the second stage of the Novel relief algorithm, four features are selected by Mahalanobis distance-based feature score and using threshold value τ =0.5.

Fig. 4 depicts the features that are selected if their relevance level is greater than or equal to τ, while the rest are discarded. The results of the relevant feature using the Novel relief algorithm are computed for $m = 475$ and $\tau = 0.5$. We observe that of all the features, only 4 features have relevance scores greater than τ. We used the averages of the 475 runs to produce the relevance score.

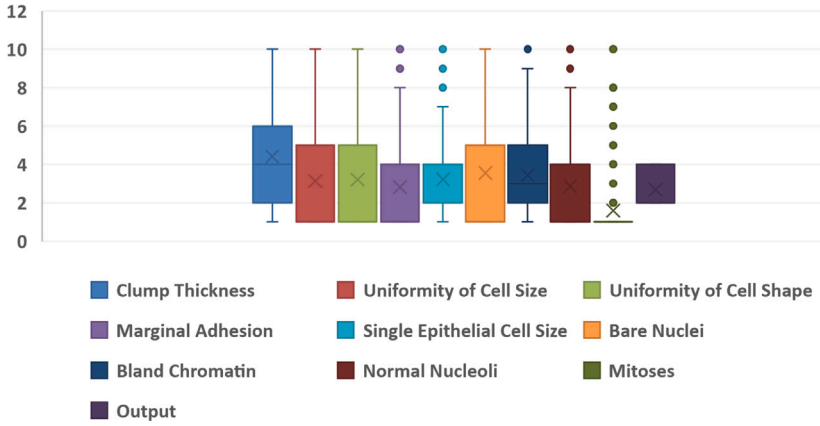


Fig. 3. Outliers of WBCD data set.

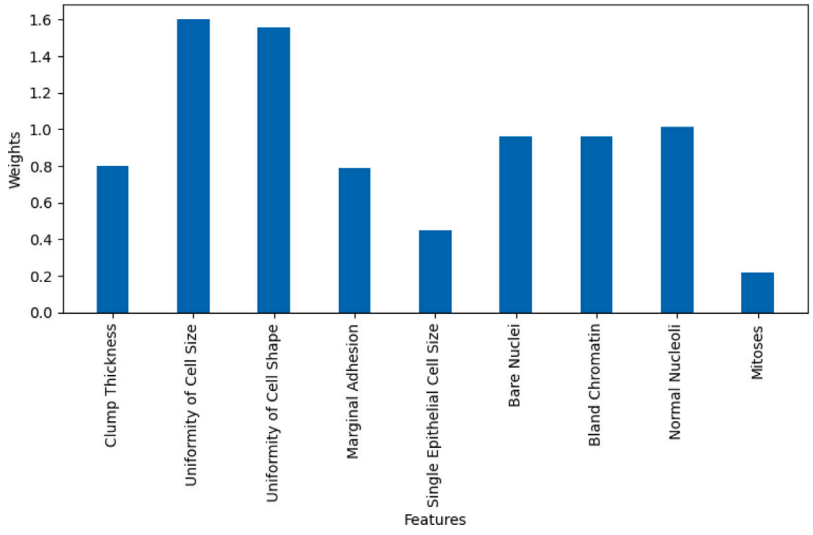


Fig. 4. Feature Selection using Novel Relief algorithm.

The advantage of using the Mahalanobis distance is that it uses group means and variance for each feature, thereby eliminating the problems of scale and correlation that are inherent in the Euclidean distance. Using this Novel algorithm, we find covariance among the features. Redundant information from the highly correlated features is also removed using this algorithm. Hence, we claim that the Novel Relief algorithm can deal with missing values, outliers and redundant features.

This algorithm also detects missing values from the data set during the feature selection process while finding the nearest hit and nearest miss using the MICE technique. Out of 699 records in the WBCD data set, 16 records contain missing feature values. For all the records, first the Novel Relief algorithm finds missing values using the MICE technique, which would be helpful during feature selection.

In the next step of the proposed model, the input–output variables are determined from the WBCD data set based on a Novel Relief algorithm. The divergence of the data from their nominal values shows the symptoms of the disease. Fuzzy sets, fuzzy domains, fuzzy rules, and fuzzy membership functions are defined using input variables.

Table 3
Range of Gaussian membership function for each selected feature.

Range of Gaussian M.F.	Feature 1	Feature 2	Feature 3	Feature 4
	[−1.214 1.983]	[−0.6995 2.5]	[−0.3437 4.904]	[−0.6332 2.521]

Table 4
Gaussian membership function with a range of Linguistic variables for each selected feature.

Membership Function (M.F.)	Feature 1	Feature 2	Feature 3	Feature 4
Low (L)	[0.6787 −1.214]	[0.6262 −0.6995]	[1.114 −0.3437]	[0.6697 −0.6332]
Medium (M)	[0.6787 0.3844]	[0.6262 0.7752]	[1.114 2.28]	[0.6697 0.9439]
High (H)	[0.6787 1.983]	[0.6262 2.25]	[1.114 4.904]	[0.6697 2.521]

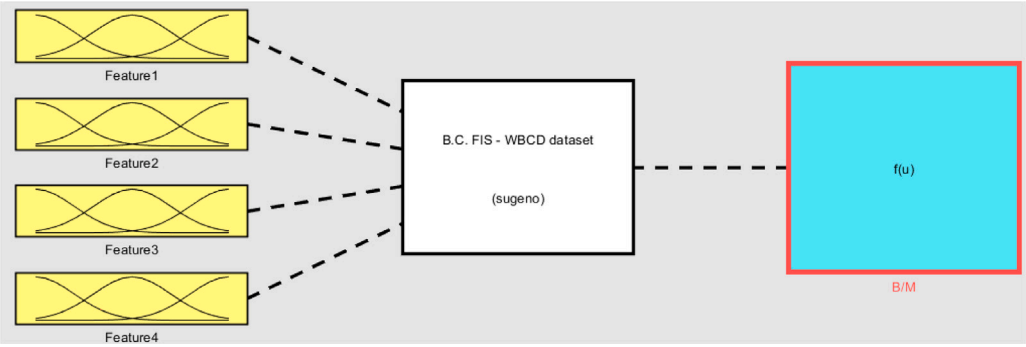


Fig. 5. Sugeno FIS for WBCD data set.

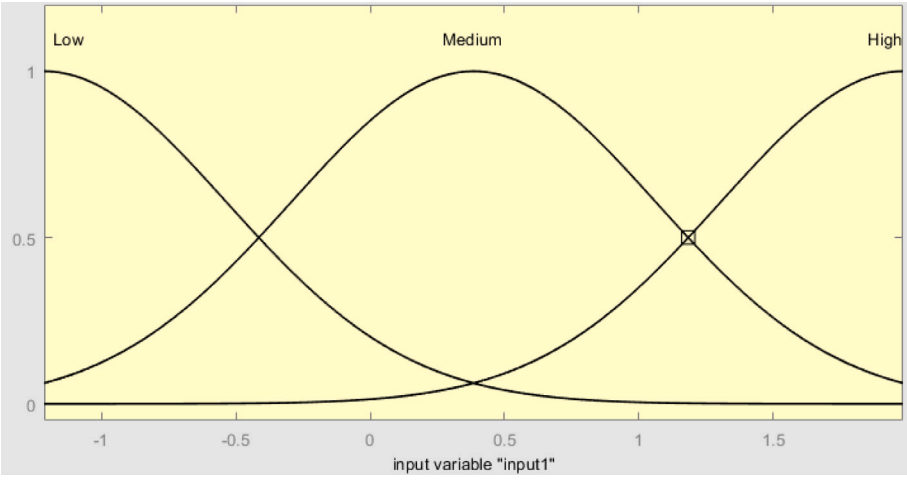


Fig. 6. Gaussian membership function of Feature 1.

All crisp input values are converted into fuzzy input values in this proposed model. We used Gaussian membership function to convert crisp values into fuzzy values, which is the first step in fuzzifying the proposed model.

Table 3 shows the range of the Gaussian membership function of the four input features.

During this fuzzification process, linguistic variables are assessed using the Gaussian membership function and they are represented by an appropriate degree of membership range from 0 to 1 using Eq. (3) and it is expressed in Table 4.

Fig. 5 is the architecture of the Sugeno Fuzzy Inference System for the WBCD data set.

Figs. 6, 7, 8 and 9 depict the Gaussian membership function for each fuzzy feature input, namely features 1, 2, 3, and 4.

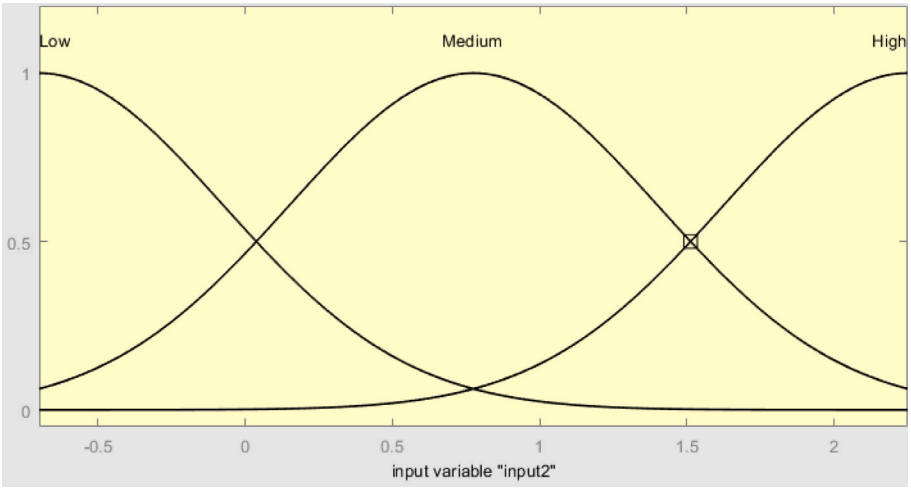


Fig. 7. Gaussian membership function of Feature 2.

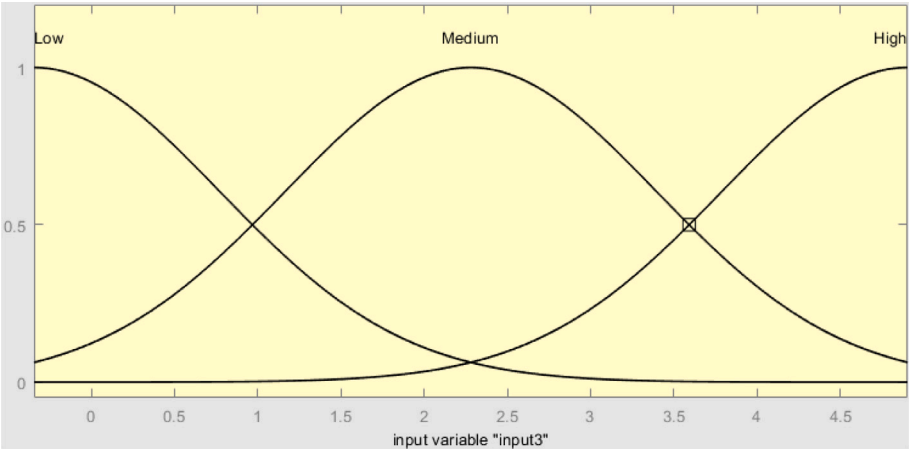


Fig. 8. Gaussian membership function of Feature 3.

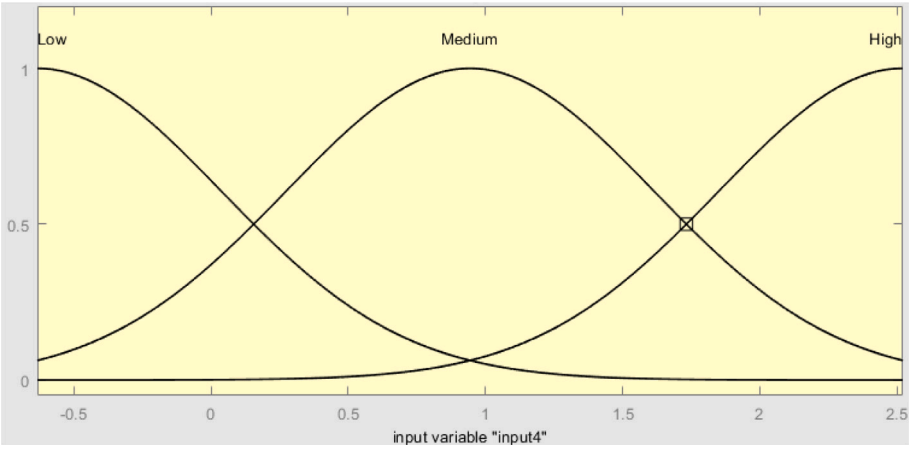


Fig. 9. Gaussian membership function of Feature 4.

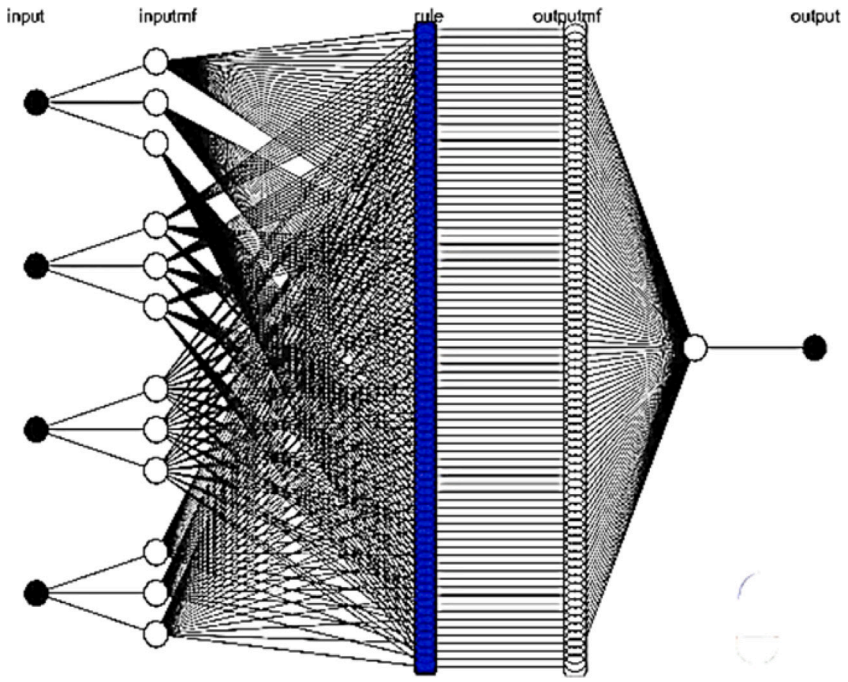


Fig. 10. Structure of proposed ANFIS model.

Fig. 10 exhibits the structure of the proposed ANFIS model, which has four feature inputs and one output, which is the ANFIS model's Neural Network representation.

4.1. Results

The model's performance and efficiency are analyzed by accuracy, sensitivity, precision and F-score using the confusion matrix and the following formulas:

$$Accuracy = \frac{Number\ of\ actual\ classified\ classes}{Total\ number\ of\ classes}$$

$$Sensitivity = \frac{Number\ of\ actual\ malignant\ classes}{Total\ number\ of\ malignant\ classes}$$

$$Sensitivity = \frac{Number\ of\ actual\ benign\ classes}{Total\ number\ of\ benign\ classes}$$

The WBCD data set is divided into 80%–20%, 70%–30% and 60%–40% train–test sets on the basic ANFIS model and the proposed hybrid ANFIS model. Figs. 11–13 demonstrate the effectiveness of a basic ANFIS model with diverse train–test sets. The performance of the Gaussian M.F. model has been further assessed using various learning rates and epochs. We attained the highest classification accuracy of 90.71% with 60%–40% train–test set for 0.01 learning rate and 100 epochs with a simple model.

Figs. 14, 15, and 16 illustrate the efficacy of a proposed hybrid ANFIS model using the train–test sets having the same ratios as a simple ANFIS model. The performance of the Gaussian M.F. model was further evaluated using different learning rates and epochs. The hybrid ANFIS model attained the highest classification accuracy of 99.30% using a 0.01 learning rate and 50 epochs.

Fig. 17 depicts the train data set's confusion matrix, which yields a classification rate of 93.20%. It reveals that 1.25% of malignant cases are misdiagnosed and 5.55% of benign cases are misdiagnosed.

Fig. 18 depicts the confusion matrix for the testing data set, demonstrating that the proposed model obtained 99.30% classification accuracy with a 0.7% misclassification rate.

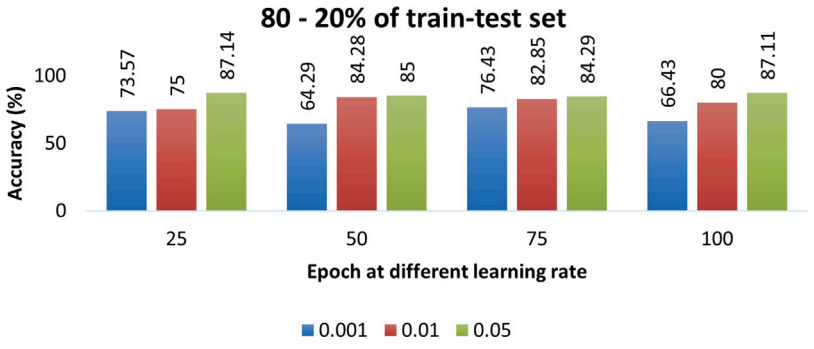


Fig. 11. Comparison of Accuracy for Simple model: different learning rate with different epoch for 80%–20% train–test set.

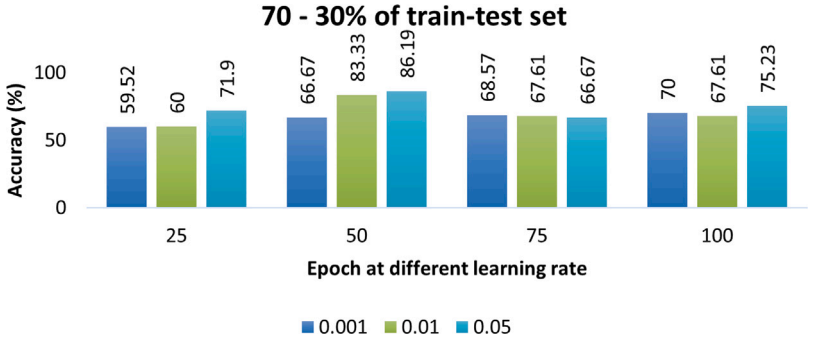


Fig. 12. Comparison of Accuracy for Simple model: different learning rate with different epoch for 70%–30% train–test set.

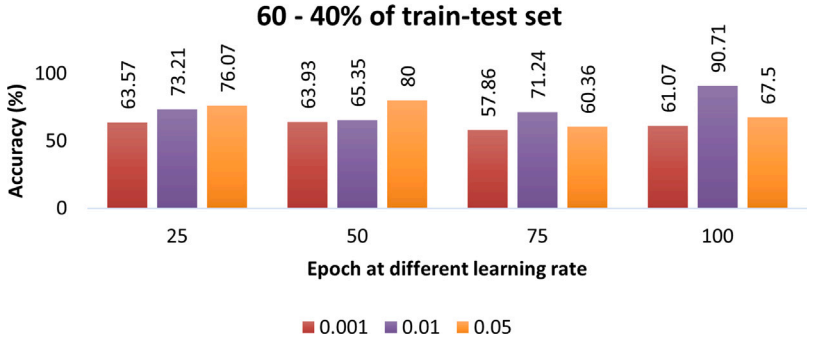


Fig. 13. Comparison of Accuracy for Simple model: different learning rate with different epoch for 60%–40% train–test set.

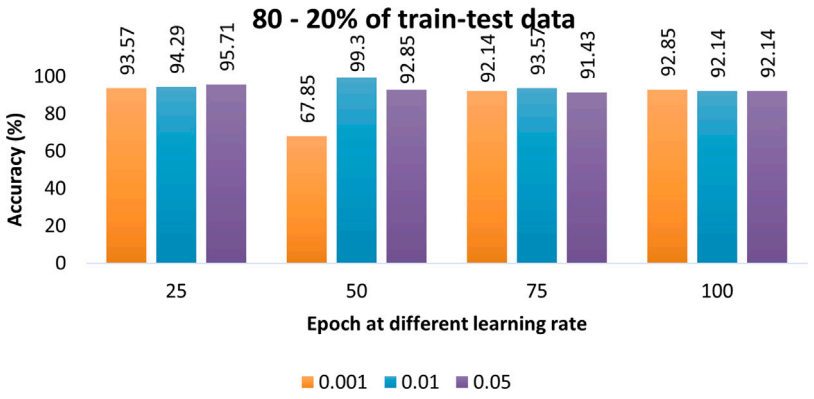


Fig. 14. Comparison of Accuracy for proposed modified model: different learning rate with different epoch for 80%–20% train–test set.

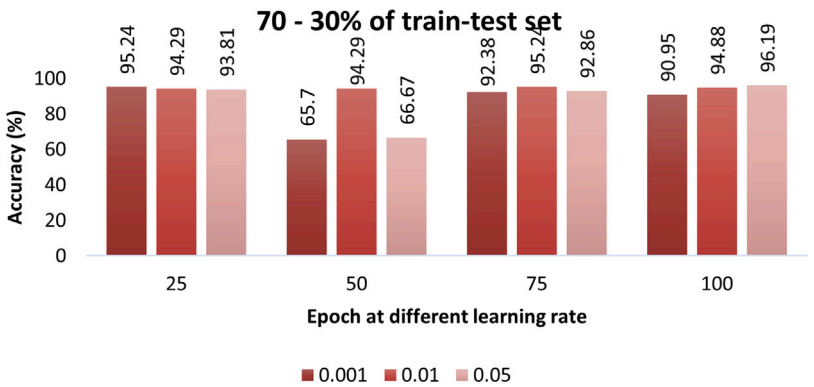


Fig. 15. Comparison of Accuracy for proposed modified model: different learning rate with different epoch for 70%–30% train–test set.

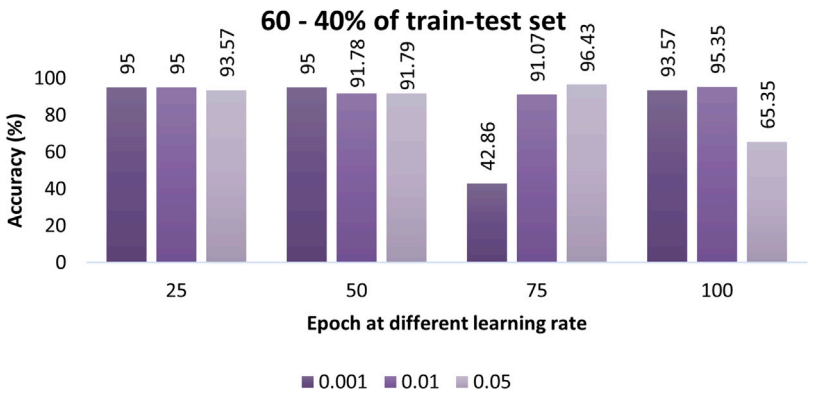


Fig. 16. Comparison of Accuracy for proposed modified model: different learning rate with different epoch for 60%–40% train–test set.

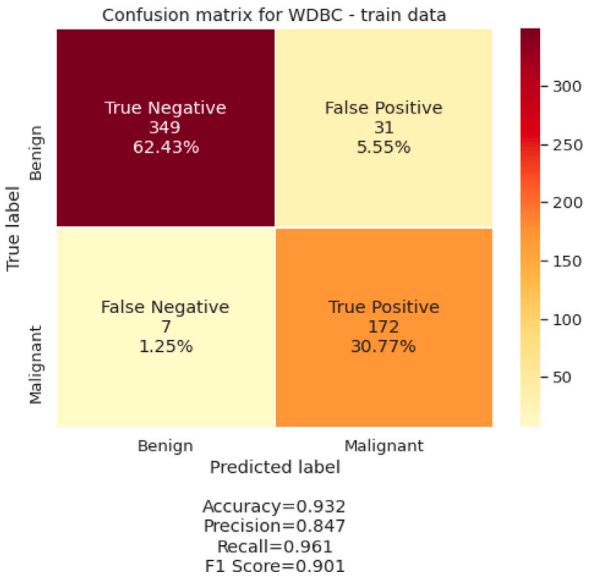


Fig. 17. Confusion matrix for Train data.

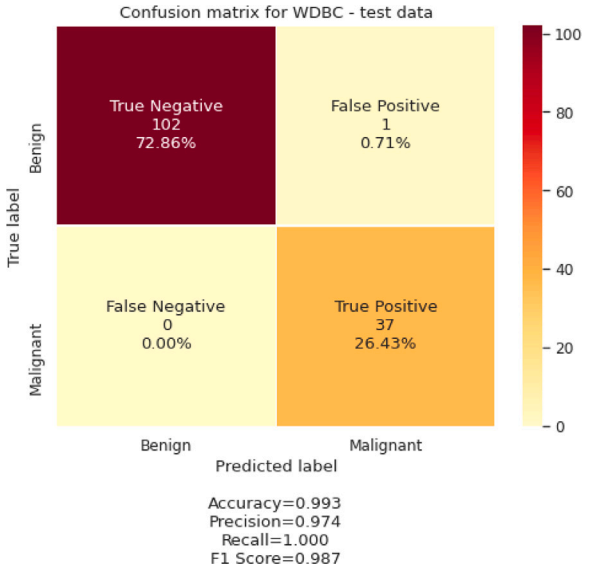


Fig. 18. Confusion matrix for Test data.

The error vs. epoch plot is shown in Fig. 19, where the error is minimized to 0.001 after a certain epoch for a learning rate of 0.001.

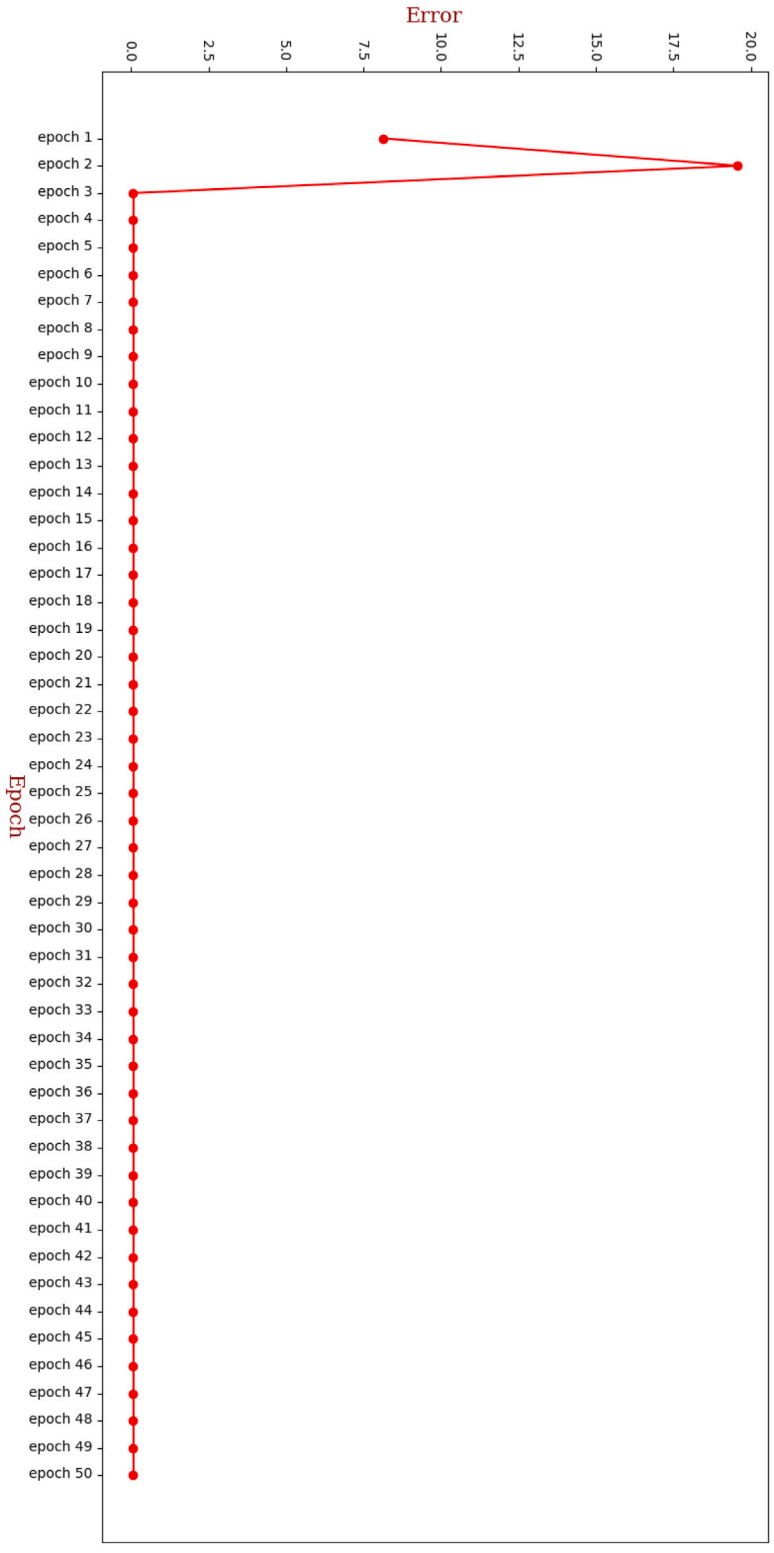


Fig. 19. Epoch vs. Error.

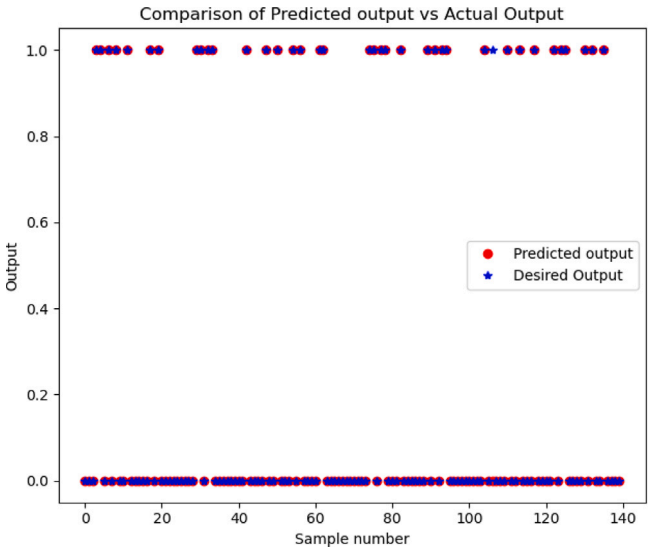


Fig. 20. Testing performance with actual output.

Table 5
Performance analysis of most popular Breast Cancer detection methods.

Authors	Year	Methodology used	Accuracy
Elif Derya Ubeyli [6]	2008	ANFIS	99.1%
Gerald Schaefer [8]	2010	Genetic algorithm+ANFIS	97.25%
Seyedeh someyeh [7]	2010	Hierarchical ANFIS	99.4%
Somayeh Naghibi [9]	2011	ANFIS	98.2%
Bekaddour Fatima [12]	2012	ANFIS	98.25%
Manisha Arrora [11]	2012	ANFIS	98.58%
Payam Zarbaksh [27]	2018	Association Rule+ANFIS	99.26%
A Sakthivel [13]	2018	ANFIS	98.92%
		SVM	93.02%
Indu Bala [15]	2019	Gravitational Search algorithm+ANFIS	96.12%
Ahmad Rizal [28]	2020	ANFIS	98.0%
Amany Mostafa [17]	2020	ANFIS	99.1%
		GFIS	97.7%
Proposed work	**2023**	**Modified Relief algorithm+ANFIS**	**99.30%**

Fig. 20 depicts the testing performance matched with the actual result. The alignment of each testing data with actual data emphasizes the testing precision.

Table 5 depicts the performance analysis and efficiency of the proposed hybrid ANFIS model to the findings of other authors working in the same domain and using the same data set.

5. Conclusion

One of the most admissible and signature tasks of medicine is disease detection. In this study, the existing Relief algorithm for feature selection is modified by eliminating outliers, imputing missing values, and using the Mahalanobis distance technique. We also propose an ANFIS model with a Novel Relief algorithm for feature selection. This proposed hybrid ANFIS model (with a Novel Relief algorithm as a feature selection technique) and the simple ANFIS model (without using a feature selection technique) have been validated on the Wisconsin Breast Cancer Data set (WBCD). The proposed hybrid ANFIS model, which has Neural Network capabilities and the Fuzzy Inference System as a rule-based system with a Novel Relief algorithm, yields excellent outcomes. The findings indicate that the hybrid approach used in the proposed ANFIS model (with a Novel Relief algorithm as a feature selection technique) gives high accuracy, sensitivity, and precision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

I have shared the link for WBCD cancer data and this data set is publicly available on UCI Machine Learning Repository.

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